

Real-Time Shuttle Tracking for U-M Hospital

Cutting customer status inquiries 40% with a public PWA serving 5K+ monthly riders

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Status: Production · 5,000+ monthly active users

Stack: Vanilla JS, Leaflet.js, Samsara API, Azure Static Web Apps, Azure Functions

TL;DR

U-M Hospital shuttle riders couldn't see where their shuttle was, so they called dispatch — every day, hundreds of times. I built a public, mobile-friendly real-time map fed by Samsara fleet telemetry through an Azure Functions proxy. Customer status inquiries dropped 40%; dispatch saved 1,095 hours/year.

How It Works

```
flowchart LR
    A["🚗 Vehicle GPS\nSamsara Fleet Telemetry"] --> B["Azure Functions Proxy\nSanitize · No PII"]
    B --> C["Public PWA\nLeaflet.js Map"]
    C --> D["📱 Rider Phone\nNo login · No install"]
    D --> E["✅ Rider sees shuttle"]
    E --> F["📞 Fewer calls to dispatch\n-40% status inquiries"]
```

The Problem

The U-M Hospital shuttle is a critical service for patients, employees, and visitors. The previous customer experience was: stand at a stop, wait, call dispatch if it took too long. Dispatch was fielding hundreds of "where's my shuttle?" calls a day — every one of them a workflow interruption that risked a worse failure elsewhere in the operation.

The Approach

- **Public-by-design.** No login, no app install. Open the link, see the map. Anything heavier loses 80% of the audience instantly.
- **Telemetry, not GPS pings.** Samsara already had high-frequency vehicle locations. I just needed to expose a sanitized slice publicly via Azure Functions — no PII, just position and route.
- **Leaflet, not Mapbox/Google.** Open-source, no per-tile cost, fast.
- **Mobile-first PWA.** Most riders are on phones; the page installs to home screen on iOS and Android.

What I'd Do Differently

- I should have added ETAs from day one — riders kept asking, "yes I see it, but *when?*" Adding that with route progress estimates next.
- The accessibility pass came late. Adding screen-reader announcements for stop arrivals.

Results

Metric	Before	After
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Monthly active riders	—	5,000+
Customer status inquiries	baseline	–40%
Dispatch hours saved	—	1,095/year

Why this matters for S&O roles

This project demonstrates the S&O motion in its simplest form: identify the real cost of a broken process, design the minimum intervention that fixes it, instrument it, and measure the result.

- **"Draw interpretable insights from sophisticated business analyses."** The 40% inquiry drop and 1,095 dispatch hours saved aren't estimates — they're measured from Samsara telemetry and dispatch log data. The number is defensible because I built the measurement layer first.
- **"Design processes, tools, and operating cadences."** The decision to make this public-by-design (no login, no install) is an operating cadence choice, not a technical one. Friction at the access point would have killed adoption and left the dispatch problem unsolved.
- **"Develop comprehensive strategies that solve complex business challenges."** The U-M Hospital account is the company's largest. A degraded rider experience is a contract-renewal risk. This tool is a retention strategy as much as an ops improvement.